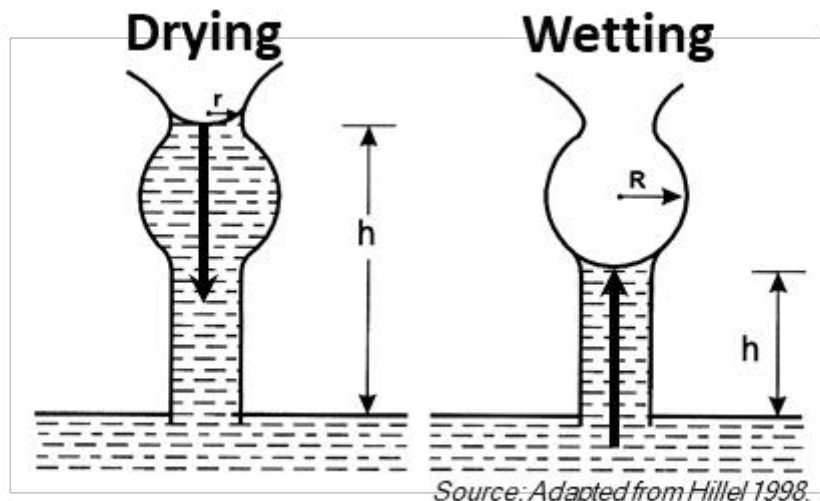




Source: Adapted from Hillel 1998.

The “ink bottle” effect describes a condition where the equilibrium height of suspended capillary water in a porous medium depends on the radius of the pore channels (r and R) and whether those pores are draining or wetting. Soil pores are geometrically heterogenous as they have variable size composition. Small radii in discontinuous pore channels govern the height of suspended water during capillary drainage (drying) while large radii limit capillary rise (wetting). Haines (1930) was the first to note the ink bottle effect as contributing toward the observed phenomenon of hysteresis in soil sorption and desorption processes.